

Cricket DRS Outcome Prediction System: A Natural Language Processing Approach to Modelling Decision Review System Outcomes

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Data Science Minor Project

April 2026

Abstract—The Decision Review System (DRS) allows cricket teams to challenge on-field umpiring decisions using ball-tracking technology. The trajectory data that technology produces—pitch coordinates, impact location, predicted path—is proprietary and closed to researchers. This project addresses that constraint by parsing ball-by-ball commentary text to reconstruct the same physical features from natural language. Those features feed a machine learning model that predicts whether a given review will succeed. The system also tracks umpire decision patterns across matches and calculates the expected value of taking a review in any match situation. Development runs across eight weeks, covering data collection and annotation, NLP feature engineering, probabilistic modelling, and deployment of a FastAPI backend with an interactive Streamlit dashboard.

Index Terms—Cricket, Decision Review System, Natural Language Processing, Machine Learning, Hawkeye, Ball-Tracking Proxy, Predictive Analytics, Expected Value, Umpire Behaviour

I. INTRODUCTION

A. Background

Cricket has been governed by a recognisable set of laws for over a century, but the *Leg Before Wicket* (LBW) dismissal remains among the most difficult to adjudicate. The on-field umpire must judge, in real time, whether a delivery that struck the batter’s pad would have gone on to hit the stumps—estimating three spatial quantities simultaneously: where the ball pitched, where it made contact with the pad, and where it was heading—under time pressure and with limited vantage.

To reduce the impact of these errors, the International Cricket Council (ICC) introduced the **Decision Review System (DRS)** in 2008 [1]. Under the DRS, each team gets a limited number of challenges per innings. When a review is called, a ball-tracking system—most commonly *Hawkeye*—rebuilds the delivery’s three-dimensional trajectory from multi-camera data and predicts whether it would have hit the stumps [2]. The result is one of three outcomes: *Hitting* (decision overturned), *Missing* (decision stands), or *Umpire’s Call* (too close to overturn; the original decision is retained) [3].

B. Problem Statement

Despite Hawkeye’s technical sophistication, its raw trajectory data is proprietary and unavailable to researchers. Analysts can observe the final verdict on broadcast, but cannot independently audit the numbers behind it, study how individual umpires perform at scale, or build data-driven tools for strategic review decisions.

This project addresses that gap by developing an *NLP-based physics proxy*—a system that extracts the same physical features Hawkeye measures directly from freely available match commentary text, and uses them to predict DRS review outcomes.

C. Objectives

The project pursues four objectives:

01. Commentary-to-Physics Mapping: Build an NLP pipeline that converts commentary sen-

tences (e.g., “Pitching in line, impact on middle, hitting leg stump”) into structured physical features.

- O2. Review Outcome Prediction:** Train and evaluate models that predict the probability of a review succeeding given those features and match context.
- O3. Umpire Behavioural Modelling:** Profile umpire tendencies—trigger rates, fatigue effects, venue patterns—to refine predictions based on who is standing.
- O4. Strategic Decision Support:** Build an expected-value engine that tells teams whether a review is worth taking given the match situation and reviews remaining.

D. Scope

The system covers international men’s and women’s cricket across Test, ODI, and T20I formats. Data collection targets matches from 2018 to 2025. The project does not aim to replace Hawkeye or any official adjudication process; it is an analytical and research tool.

E. Organisation of the Report

Section II surveys the relevant literature. Section III covers the system architecture and methodology. Section IV describes the chosen technology stack. Section V lays out the implementation plan. Section VI covers evaluation. Section VII shows the project schedule. Section VIII identifies risks and mitigations. Section IX lists expected contributions. Section X describes how this could extend into a full-match analysis platform. Section XI concludes.

II. LITERATURE REVIEW

A. Cricket Analytics and Data Science

Quantitative cricket research has been active for well over a decade. Early work covered player performance metrics and win probability models [4], [5]. More recent studies have applied machine learning to match outcome prediction [6]–[8], batsman-bowler matchups [9], and batting order optimisation. The DRS has received little attention in this literature, and using raw commentary text as a primary feature source for DRS prediction remains largely unexplored.

B. DRS-Specific Research

Research specifically on the DRS is thin. Most published analyses look at aggregate statistics—success rates by team, format, or dismissal type—rather than attempting prediction [10]. The central obstacle is the same one this project faces: Hawkeye data is not publicly available. Studies that do attempt to model LBW decisions rely on small, manually curated datasets, which constrains model complexity and limits how far results generalise.

C. NLP in Sports Analytics

NLP has been applied to sports in a few useful directions. Social media sentiment has been used to track momentum during live matches [11]. Commentary mining for event detection and summarisation has been explored in football [12]. In cricket, commentary text has been parsed for automatic scoreboard updates and event classification [13]. None of these works attempts to reconstruct physical delivery features from text—the text-to-physics mapping that forms the core of this project.

D. Identified Research Gaps

The literature leaves four clear gaps:

- No existing system reconstructs ball-tracking features from commentary text for DRS prediction.
- Umpire behavioural modelling at scale—across hundreds of matches and many umpires—has not been done.
- The strategic question of when to review has received limited formal treatment using decision theory.
- There is no publicly available annotated corpus of DRS review events with associated commentary text.

III. PROPOSED METHODOLOGY

A. System Architecture Overview

The proposed system comprises four layers—data acquisition, feature engineering, modelling, and application—as illustrated in Figure 1.

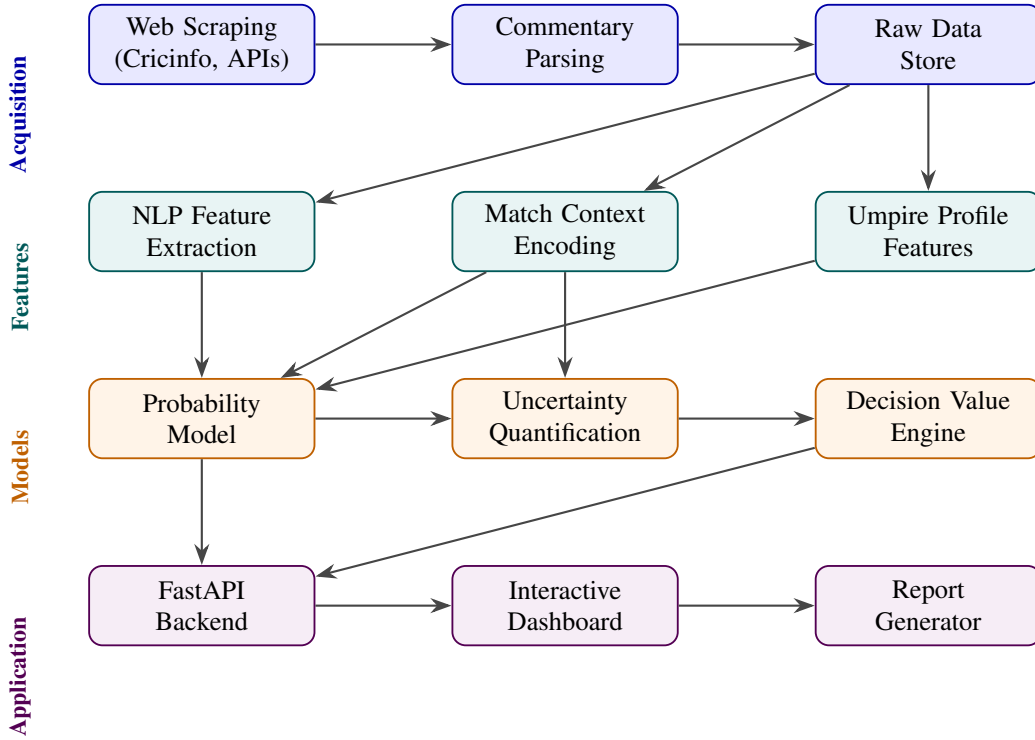


Fig. 1. End-to-end system architecture of the proposed Cricket DRS Outcome Prediction System, showing data flow across the four principal layers: Data Acquisition (blue), Feature Engineering (teal), Modelling (orange), and Application (violet).

B. Data Acquisition Layer

Data is collected from three sources:

- **Ball-by-ball datasets:** Records from ESPN Cricinfo and Cricsheet [14] giving delivery-level metadata—bowler, batsman, runs, wicket type, over number.
- **Commentary feeds:** The natural language description that accompanies each delivery, sourced from Cricinfo’s live commentary.
- **Match metadata:** Venue, date, format, playing XI, toss result, and match outcome.

A scraping pipeline links each DRS event to its commentary text and stores the combined record in PostgreSQL. All datasets are versioned with DVC.

C. Feature Engineering and NLP Layer

Feature extraction from commentary text proceeds in three stages.

1) *Rule-Based Physical Feature Extraction:* A set of regular expressions and keyword dictionaries picks out explicit spatial references. The sentence “Pitching outside off, hitting middle and leg” maps directly to `pitch_line = outside_off`,

`impact_zone = middle_and_leg`, `hitting = True`. These rules are fast, interpretable, and serve as the baseline.

2) *Statistical NLP Feature Extraction:* For commentary that is ambiguous or implicit, two approaches are evaluated:

- **TF-IDF + Classical ML:** TF-IDF vectorisation of commentary text fed into a logistic regression or SVM classifier. Lightweight and interpretable.
- **Transformer-Based Encoding:** Fine-tuning BERT or RoBERTa on the annotated corpus for sequence classification across pitch location, impact zone, and trajectory. This approach handles contextual and implicit language more accurately, though it requires more training data and computational resources.

3) *Commentator Certainty Scoring:* A separate model scores how confident the commentator sounds. “Clearly missing” and “looked plumb” sit at opposite ends of the certainty spectrum. This score $c \in [0, 1]$ is included as an additional input feature.

D. Match Context Features

The model also takes structured match context:

- **Format:** Test, ODI, or T20I—each has different review norms and risk dynamics.
- **Match situation:** Over number, innings, wickets fallen, run rate, required run rate.
- **Bowling type:** Pace or spin, over or around the wicket.
- **Review state:** Reviews remaining for the challenging team.

E. Modelling Layer

1) *Primary Prediction Model:* The core model predicts the probability of a review succeeding:

$$\hat{p} = f(\mathbf{x}_{\text{NLP}}, \mathbf{x}_{\text{context}}, \mathbf{x}_{\text{umpire}}) \quad (1)$$

where $\mathbf{x}_{\text{NLP}}$ is the NLP feature vector, $\mathbf{x}_{\text{context}}$ is match context, and $\mathbf{x}_{\text{umpire}}$ is the umpire profile. Candidate model families are gradient-boosted trees (XGBoost, LightGBM), logistic regression as a baseline, and neural classifiers.

2) *Uncertainty Quantification:* Marginal calls—those near the Hitting/Umpire’s Call boundary—need more than a point estimate. Two approaches handle this: Platt scaling or isotonic regression to calibrate probabilities, and optionally a Bayesian logistic regression or Gaussian process model to produce full posterior distributions with credible intervals.

3) *Strategic Decision Value Engine:* The expected gain from taking a review is:

$$\text{EV}_{\text{review}} = P(\text{success} \mid \mathbf{x}) \times V_{\text{wicket}} - C_{\text{review}} \quad (2)$$

where V_{wicket} is the wicket’s value in the current match context (estimated via Win Probability Added) and C_{review} is the cost of spending a review. A review is recommended when $\text{EV}_{\text{review}} > 0$.

4) *Umpire Behavioural Model:* Umpire profiles are built from historical data and cover:

- **Trigger rate:** How often the umpire gives LBW decisions.
- **Overturn rate:** The fraction of their decisions reversed on review.
- **Fatigue model:** Whether accuracy drops session-by-session across a day’s play.
- **Venue/team bias:** Statistical tests for whether decision patterns shift by ground or opposition.

F. Application Layer

The system exposes two interfaces:

- **REST API (FastAPI):** Takes commentary text and match context, returns success probability, recommended action, and umpire profile.
- **Interactive Dashboard:** A Streamlit interface showing review outcome heatmaps, umpire tendency charts, success-rate breakdowns, and a “Should You Review?” calculator.

IV. TECHNOLOGY STACK

All major candidates were evaluated across ML frameworks, NLP libraries, web layers, databases, visualisation tools, and Bayesian inference libraries. The full trade-off analysis is in the companion document *Technology Stack Comparisons*. The chosen stack:

- **Language:** Python 3.11+
- **Data Processing:** Pandas, NumPy, DVC (data version control)
- **NLP:** spaCy (rule-based extraction and tokenisation), HuggingFace Transformers (fine-tuning), custom regex
- **Machine Learning:** Scikit-learn (baseline and preprocessing), XGBoost and LightGBM (primary models)
- **Bayesian Modelling:** PyMC (uncertainty quantification)
- **Web Framework:** FastAPI (REST backend), Streamlit (interactive dashboard)
- **Database:** PostgreSQL (production), SQLite (development and testing)
- **Experiment Tracking:** MLflow
- **Testing:** pytest, Great Expectations (data pipeline validation)
- **Deployment:** Docker, Vercel

V. IMPLEMENTATION PLAN

Development is structured across four phases.

A. Phase 1: Data Acquisition and Annotation

- 1) Develop scraping scripts for ESPN Cricinfo commentary and Cricsheet ball-by-ball data.
- 2) Build an ingestion pipeline that links commentary text to delivery-level metadata.
- 3) Identify all DRS review events in the dataset (targeting 500+ reviews across formats).

- 4) Manually annotate a subset (200+ reviews) with structured labels: pitch line, impact zone, predicted trajectory, outcome (Overturn / Umpire's Call / Retained).
- 5) Set up PostgreSQL schema and DVC for data versioning.

B. Phase 2: NLP Feature Engineering

- 1) Implement rule-based feature extraction using spaCy Matcher and regex patterns.
- 2) Train a TF-IDF + logistic regression baseline for commentary classification.
- 3) Fine-tune BERT or RoBERTa on the annotated corpus for sequence classification (pitch location, impact zone, trajectory).
- 4) Develop the commentator certainty scoring module.
- 5) Validate extraction accuracy against annotated ground truth.

C. Phase 3: Modelling and Evaluation

- 1) Train XGBoost and LightGBM models on extracted features and match context.
- 2) Implement probability calibration (Platt scaling, isotonic regression).
- 3) Build the Bayesian uncertainty model using PyMC.
- 4) Develop the umpire behavioural profiling module.
- 5) Implement the decision value engine (Equation 2).
- 6) Run ablation studies to measure each feature group's contribution.

D. Phase 4: Application Layer and Deployment

- 1) Build the FastAPI backend with prediction, umpire profile, and EV endpoints.
- 2) Develop the Streamlit dashboard with:
 - Review outcome heatmaps (by pitch location, impact zone).
 - Umpire tendency visualisations.
 - "Should You Review?" interactive calculator.
 - Historical success rate breakdowns by team, format, and venue.
- 3) Write unit and integration tests.
- 4) Containerise with Docker.
- 5) Prepare documentation and user guide.

VI. EVALUATION STRATEGY

A. NLP Extraction Evaluation

The extraction pipeline is validated against manual annotations using:

- **Precision, Recall, F1-Score:** Per-feature (pitch line, impact zone, trajectory) and macro-averaged.
- **Exact Match Ratio:** Fraction of deliveries where all extracted features match the ground truth.

B. Prediction Model Evaluation

- **AUC-ROC:** Primary metric; measures discriminative power across all thresholds.
- **Brier Score:** Measures probability calibration—how close predicted probabilities are to observed outcomes.
- **Log Loss:** Penalises confident wrong predictions; appropriate for probabilistic output.
- **Precision-Recall at operational thresholds:** Relevant because successful reviews are the minority class.

C. Baseline Comparisons

Four baselines set the performance floor:

- **Random baseline:** Predicts success at the base rate.
- **Majority class baseline:** Always predicts the most common outcome.
- **Rule-only baseline:** NLP features only, no statistical model.
- **Context-only baseline:** Match context only, no NLP.

D. Validation Methodology

Models use stratified 5-fold cross-validation, stratified by format and DRS outcome to keep class balance across folds. Final evaluation uses a temporal holdout—the most recent matches—to test how models perform on data they have not seen.

VII. PROJECT TIMELINE

Figure 2 shows the eight-week schedule. Phases overlap at their boundaries so earlier outputs feed directly into later work without idle time between them.

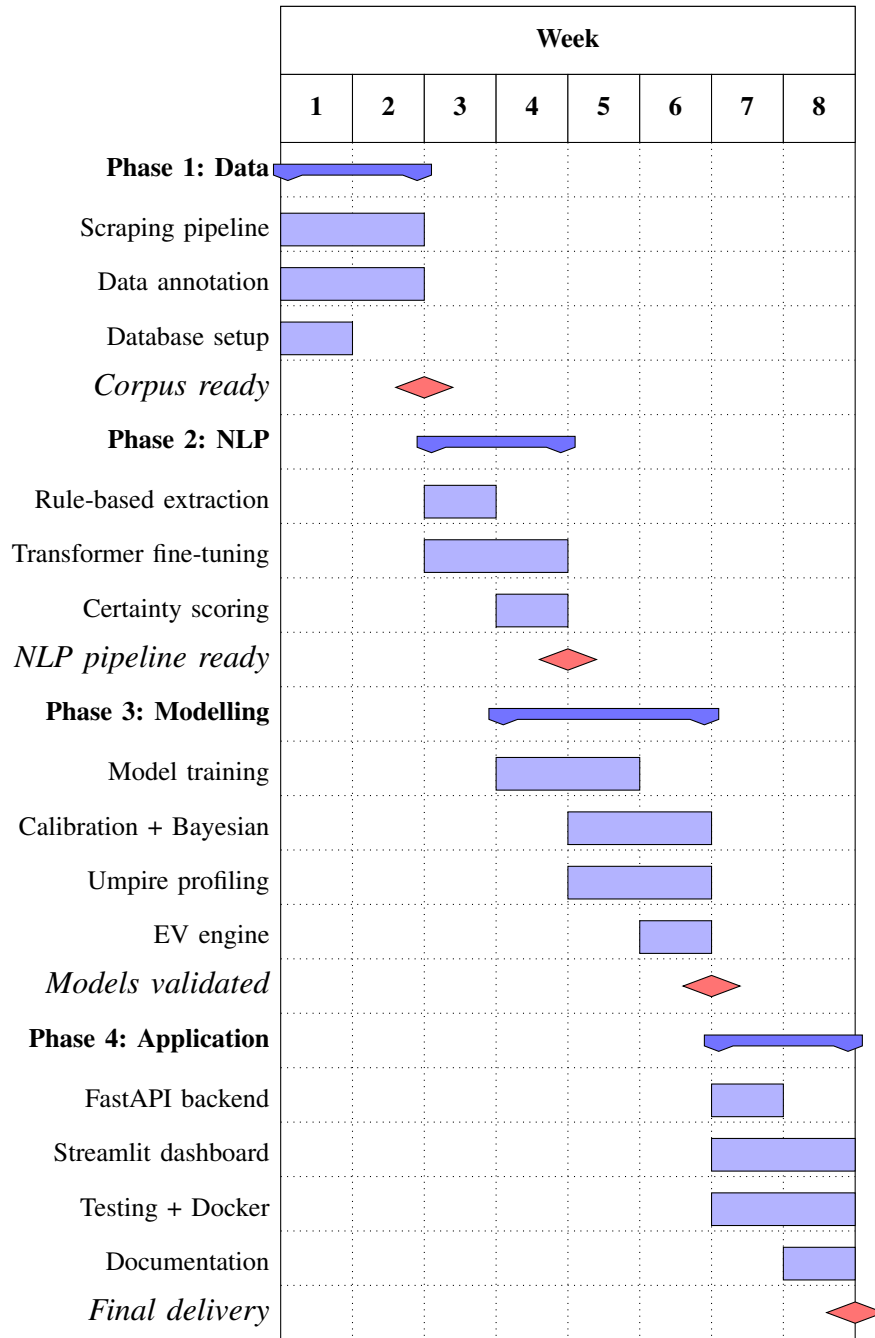


Fig. 2. Gantt chart illustrating the eight-week project development timeline across four phases.

VIII. RISK ANALYSIS AND MITIGATION

Table I lists the main risks and mitigations.

IX. EXPECTED OUTCOMES AND CONTRIBUTIONS

The project should produce five concrete outputs:

C1. Novel Methodology: An open-source text-to-physics pipeline for extracting ball-tracking

features from cricket commentary—to our knowledge, the first of its kind.

C2. Predictive Model: A calibrated model predicting DRS review success, targeting AUC-ROC ≥ 0.75 .

C3. Umpire Profiles: Quantitative behavioural profiles for active international umpires, surfacing patterns that aggregate statistics hide.

TABLE I
RISK ASSESSMENT MATRIX

Risk	Severity	Mitigation Strategy
Commentary data is inconsistent across sources	High	Standardise through a preprocessing normalisation layer; cross-validate labels against multiple independent sources.
Insufficient DRS review samples for training	High	Augment with synthetic paraphrases via back-translation; leverage transfer learning from general cricket commentary corpora.
NLP extraction accuracy is too low for reliable prediction	Medium	Fall back to rule-based features as a hard floor; ensemble rule-based and statistical approaches with confidence-weighted fusion.
Transformer fine-tuning requires GPU resources	Medium	Use Google Colab Pro or university HPC cluster; constrain model size to DistilBERT to reduce memory footprint by 40%.
Web scraping blocked or rate-limited	Medium	Respect <code>robots.txt</code> ; implement request caching and exponential back-off; use Cricsheet as a scraping-free fallback [14].
Class imbalance (fewer successful reviews)	Medium	Apply SMOTE oversampling or class-weighted loss; use AUC-ROC and Brier Score as primary metrics rather than raw accuracy.
Scope creep beyond schedule	Low	Enforce strict phase-gate reviews; defer transformer-based NLP and umpire profiling to v2 if timeline is threatened.

C4. Decision Support Tool: A dashboard and API that lets analysts, broadcasters, and coaching staff evaluate review decisions with actual data.

C5. Reproducibility: All code, pipelines, and trained models version-controlled and documented.

X. FUTURE WORK

The current project scope is deliberately narrow, but the underlying architecture is not. The same NLP infrastructure applies to a broader problem: analysing a complete cricket match from commentary text alone.

A. Full-Match Commentary Analysis

The NLP pipeline is not inherently specific to DRS events. By extending the feature vocabulary to cover scoring shots, bowling variations, and fielding positions, the system could produce a structured, machine-readable innings representation from raw commentary—an open-data alternative to proprietary ball-tracking feeds for general analysis.

B. Real-Time Win Probability and Momentum Modelling

With per-delivery NLP features available across a full match, a live win probability model becomes tractable. Commentary sentiment and certainty signals—already captured by the certainty

scoring module—could serve as indicators of momentum in ways that pure run-rate models cannot, capturing events such as dropped catches, close decisions, or sustained pressure periods.

C. Batsman and Bowler Profiling

The umpire behavioural profiling module can be adapted to construct analogous profiles for batters and bowlers. The system could model a batter’s susceptibility to specific delivery types by correlating NLP-extracted bowling features with dismissal events, or identify a bowler’s most effective variations across different match situations.

D. Multi-Format Tactical Analysis

T20I powerplay bowling and Test match fifth-day field placement are fundamentally different strategic problems. At scale, commentary mining across hundreds of matches can surface format-specific tactical patterns without requiring manual annotation.

E. Integration with Video and Broadcast Systems

In the longer term, NLP-derived features could be fused with computer vision models applied to broadcast footage. Commentary text provides semantic context while video provides spatial precision; together, they approach the information content of proprietary ball-tracking systems at significantly lower cost.

F. Crowd-Sourced Annotation and Community Platform

Scaling the annotated dataset beyond a small team requires broader participation. A future version of the platform could offer a community annotation interface where cricket enthusiasts contribute structured labels for DRS events, expanding the training corpus while building a user base around the tool.

XI. CONCLUSION

Cricket's DRS generates some of the richest real-time decision data in sport, and almost none of it is accessible for independent study. This project finds a way in through the one source that is freely available: match commentary. By extracting delivery features from text and training a calibrated prediction model on them, the system estimates review success probability without access to a single frame of Hawkeye data.

The four-layer architecture—from data acquisition through to the interactive dashboard—is modular by design. The umpire profiling module and expected-value engine add analytical depth that goes well beyond a basic binary classifier. The chosen stack of spaCy, HuggingFace Transformers, XGBoost, LightGBM, PyMC, FastAPI, and Streamlit is well-suited to an eight-week development scope and leaves clear paths for future extension.

Every delivery in a cricket match has commentary attached to it. That commentary is largely untapped as a structured data source. The research questions this project opens—full-match analysis, live win probability, player profiling—are arguably as significant as the prediction problem it directly addresses.

REFERENCES

- [1] International Cricket Council, "Decision Review System (DRS) Playing Conditions," ICC, 2023. [Online]. Available: <https://www.icc-cricket.com>
- [2] P. Hawkins, "Hawk-Eye: The Invention That Changed Sport," in *Engineering and Technology Magazine*, vol. 10, no. 4, pp. 52–55, 2015.
- [3] International Cricket Council, "Standard Playing Conditions for Test Matches, ODIs and T20Is," ICC, 2024.
- [4] A. J. Lewis, "Towards Fairer Measures of Player Performance in One-Day Cricket," *Journal of the Operational Research Society*, vol. 56, no. 7, pp. 804–815, 2005.
- [5] M. J. Bailey and S. R. Clarke, "Predicting the Match Outcome in One Day International Cricket Matches, While the Match is in Progress," *Journal of Sports Science and Medicine*, vol. 5, no. 4, pp. 480–487, 2006.
- [6] A. Bandulasiri, "Predicting the Winner in One Day International Cricket," *Journal of Mathematical Sciences and Mathematics Education*, vol. 3, no. 1, pp. 6–17, 2008.
- [7] V. V. Sankaranarayanan, J. Sattar, and L. V. Lakshmanan, "Auto-play: A Data Mining Approach to ODI Cricket Simulation and Prediction," in *Proc. SIAM International Conference on Data Mining*, Philadelphia, PA, 2014, pp. 1064–1072.
- [8] S. Kampakis and W. Thomas, "Using Machine Learning to Predict the Outcome of English County Twenty Over Cricket Matches," *arXiv preprint arXiv:1511.05837*, 2015.
- [9] H. H. Lemmer, "The Combined Bowling Rate as a Measure of Bowling Performance in Cricket," *South African Journal for Research in Sport, Physical Education and Recreation*, vol. 24, no. 2, pp. 37–44, 2011.
- [10] V. K. Borooah, "Upstairs and Downstairs: The Imperfections of Cricket's Decision Review System," *Journal of Sports Economics*, vol. 17, no. 1, pp. 64–85, 2016.
- [11] Y. Yu and X. Wang, "World Cup 2014 in the Twitter World: A Big Data Analysis of Sentiments in US Sports Fans' Tweets," *Computers in Human Behavior*, vol. 48, pp. 392–400, 2015.
- [12] W. van Atteveldt, M. Trilling, and C. Arcila Calderon, *Computational Analysis of Communication*. Wiley-Blackwell, 2021.
- [13] P. Rao and S. Dey, "Automatic Cricket Commentary Generation from Ball-by-Ball Data," in *Proc. International Conference on Natural Language Processing*, Patna, India, 2018, pp. 214–223.
- [14] S. Lister, "Cricsheet: Free and Open Cricket Data," 2024. [Online]. Available: <https://cricsheet.org>